

Combining Personalized Functional Mapping with Connectome-Based Predictive Modeling to Characterize the “Exposome Network” in Adolescents from the ABCD Study

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Paper:

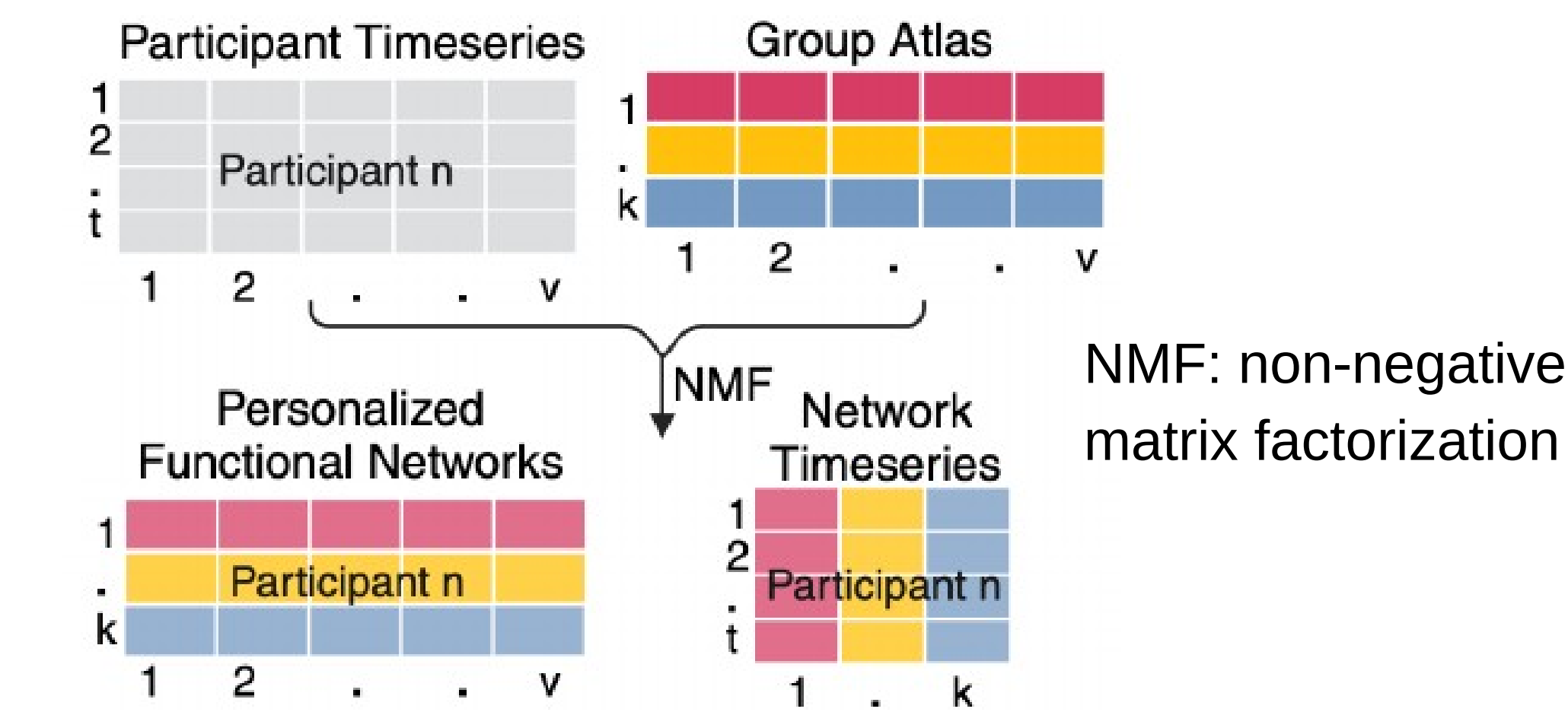


Background & Motivation

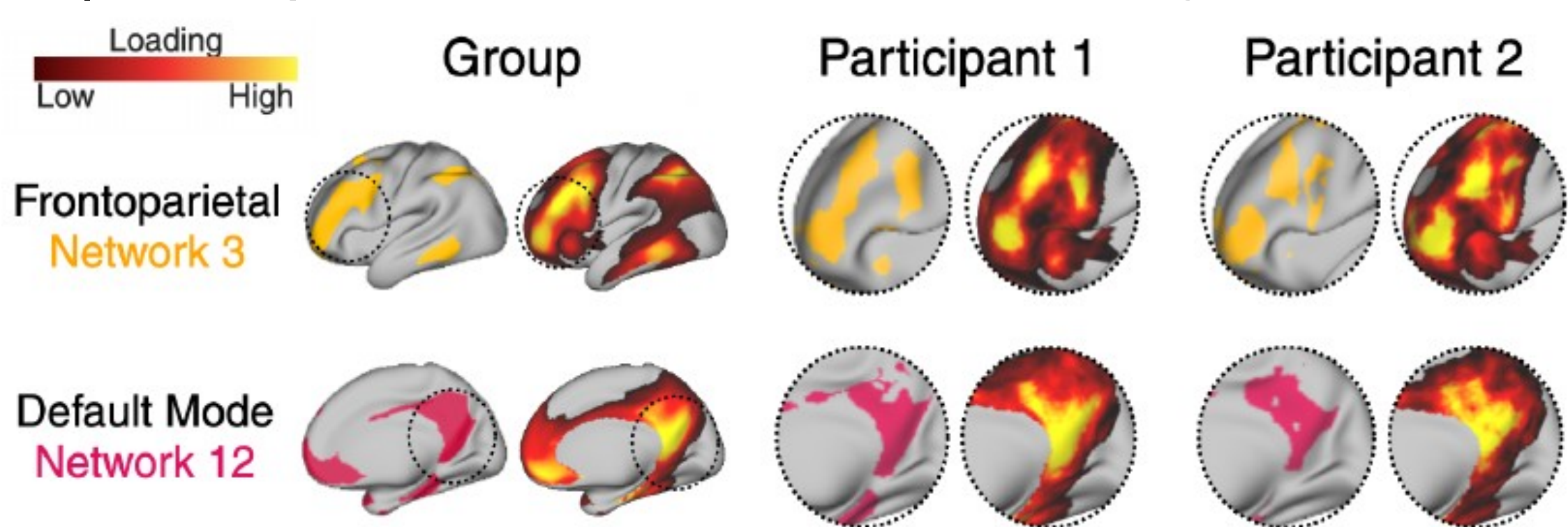
- Functional brain networks are highly heterogeneous across individuals; however, commonly used group-averaged atlases may misattribute network assignments for a given person, hindering generalizability, mechanistic interpretations, and clinical utility.
- Adolescence is a critical period of neurodevelopment of heightened brain plasticity¹ when functional brain networks undergoing protracted maturation are particularly susceptible to the collective multivariate effects of experiences, exposures, and environments (the “**exposome**”)².
- Given that adolescent neurodevelopment is associated with adult outcomes (e.g., cognition³, psychiatric risk⁴), we must robustly characterize the relationships between functional connections and phenotypes of interest.
- Connectome-based predictive modeling (CPM)** is a whole-brain, data-driven machine learning approach used to identify brain connections most relevant for predicting a given phenotype⁵.
- Interpreting CPM-derived networks typically relies on group-averaged atlases to label predictive edges and assumes homogeneous functional neuroanatomy, despite high inter-individual variability of functional brain networks⁶.
- Personalized functional networks (PFNs)** overcome this challenge by accounting for individual differences in the size, shape, and spatial layout of brain networks.

Personalized Connectome-Based Predictive Modeling (pCPM)

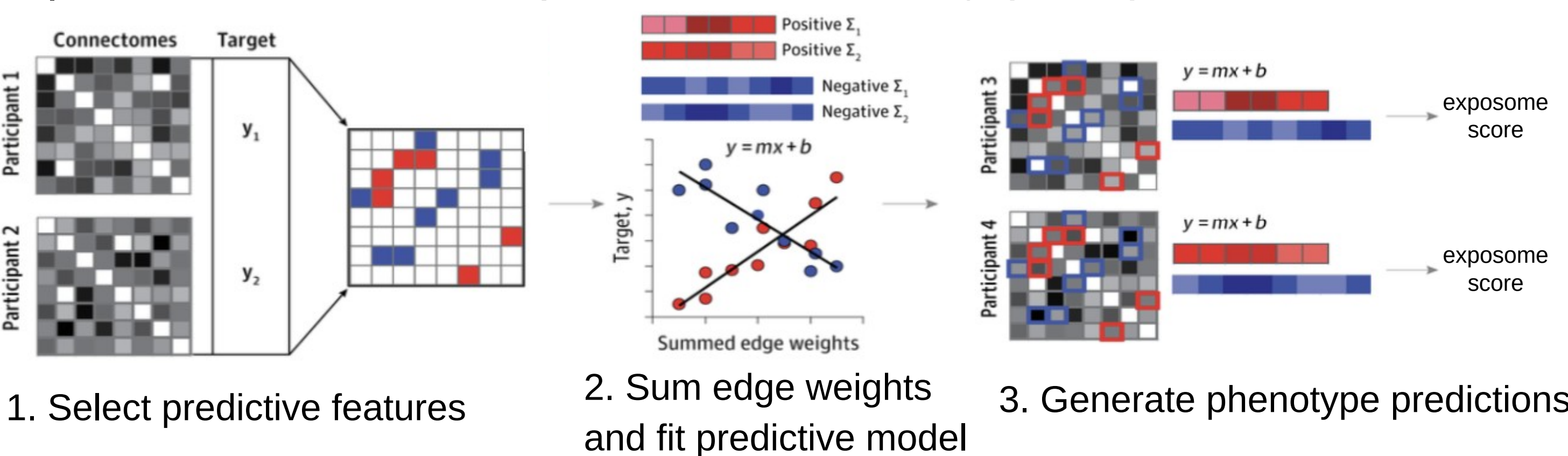
a | Deriving personalized functional networks (PFNs)



b | Example PFNs from the ABCD Study



c | Connectome-based predictive modeling (CPM)

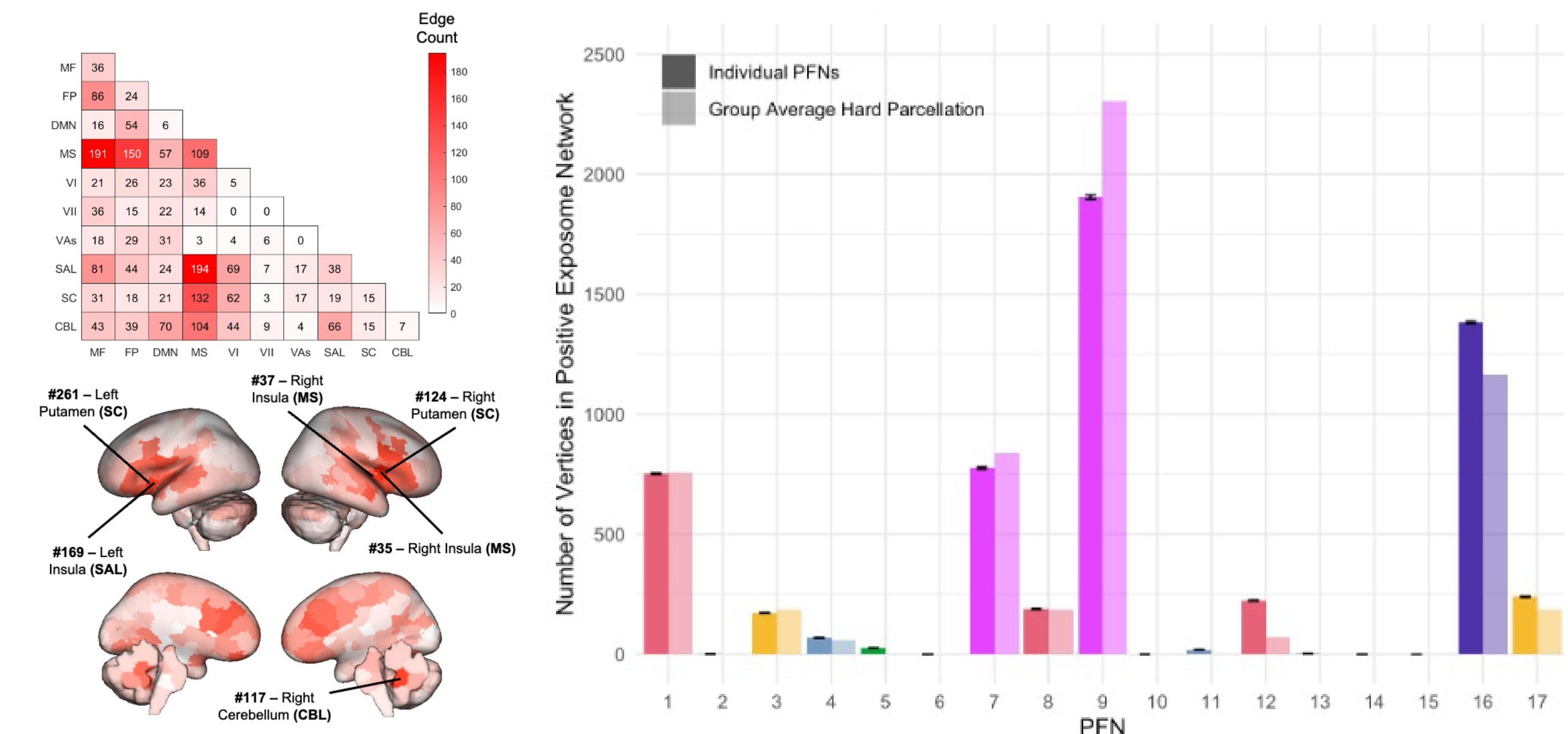


Adapted from Keller et al., 2025 (*British Journal of Psychiatry*) and Lichenstein et al., 2026 (*Developmental Cognitive Neuroscience*)

- Here, we introduce **personalized CPM (“pCPM”)**, which integrates PFNs into a CPM framework⁷.
- As a proof-of-concept test case, we applied pCPM to identify connections most strongly associated with the exposome to compare an individualized vs. a group-average approach.
- We used person-specific brain atlases of 17 PFNs⁸ previously derived⁹ from the ABCD Study¹⁰ (n=6,972, ages 9-10) across 3 task-based and 1 resting-state fMRI scans.
- We applied CPM with 10-fold cross-validation (90% training data, 10% testing data), controlling for age, sex, and head motion, to predict exposome scores¹¹ from whole-brain connectivity.
- We mapped the top 10% of highest degree nodes of the “**positive exposome network**”—the set of functional edges whose connectivity was positively associated with individuals’ exposome scores—onto both the (group-averaged) Shen atlas and (individualized) PFNs.

Functional system contributions to “exposome network” differ by group vs. personalized approach

- Mapping the exposome network back to a group-averaged atlas revealed overlap with the motor sensory, medial frontal, and salience networks.
- Mapping the exposome network back to individualized PFNs revealed overlap with the auditory (PFN 16), ventral attention (PFNs 9 and 7), and default mode (PFN 1) networks.
- The differences in implicated networks between group-average and individualized atlases underscore how **atlas choice meaningfully changes which functional networks are implicated in brain-phenotype associations**.



Lichenstein et al., 2026 (*Developmental Cognitive Neuroscience*)

Discussion

- These findings demonstrate how leveraging personalized neuroscience may inform research into individual differences more broadly.
- As group-averaged atlases obscure meaningful functional heterogeneity across individuals, large-scale brain networks identified using group-average approaches may yield inaccurate characterizations of implicated networks in brain-phenotype associations.
- pCPM seeks to facilitate more precise interpretations by integrating each individual’s unique brain organization, enhancing network-level interpretations of brain-phenotype findings.

Future Directions

Although we used the exposome to demonstrate the utility of pCPM, this technique can be broadly applied to any phenotype of interest (e.g., cognitive functioning, motivated behaviors, transdiagnostic psychiatric features) to advance our understanding of whole-brain functional connectivity patterns underlying individual differences in cognitive neurodevelopment.

References

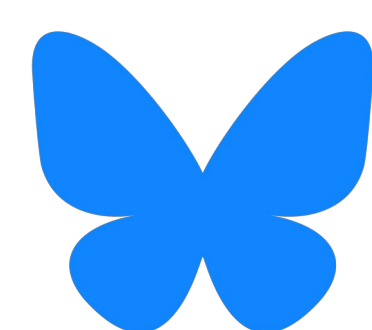
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Acknowledgments

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